

Learning Rate vs Convergence

Check browser console

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```
script.js:13
Training with learning rate = 0.001
Live reload enabled. index.html:39
LR 0.001 | Epoch 0 | Loss 583.5269 script.js:20
LR 0.001 | Epoch 20 | Loss 235.2257 script.js:20
LR 0.001 | Epoch 40 | Loss 101.4250 script.js:20
LR 0.001 | Epoch 60 | Loss 49.9336 script.js:20
LR 0.001 | Epoch 80 | Loss 30.0276 script.js:20
script.js:13
Training with learning rate = 0.01
LR 0.01 | Epoch 0 | Loss 969.0512 script.js:20
LR 0.01 | Epoch 20 | Loss 14.6848 script.js:20
LR 0.01 | Epoch 40 | Loss 12.8074 script.js:20
LR 0.01 | Epoch 60 | Loss 11.1848 script.js:20
LR 0.01 | Epoch 80 | Loss 9.7677 script.js:20
script.js:13
Training with learning rate = 0.1
LR 0.1 | Epoch 0 | Loss 819.1730 script.js:20
LR 0.1 | Epoch 20 | Loss 211082512.0000 script.js:20
LR 0.1 | Epoch 40 | Loss 55577845694464.0000 script.js:20
LR 0.1 | Epoch 60 | Loss 14633615758349828096.0000 script.js:20
LR 0.1 | Epoch 80 | Loss 3.853024469064926e+24 script.js:20
> |
```

Assignment 1: Linear Regression using TFJS

Open browser console to view output

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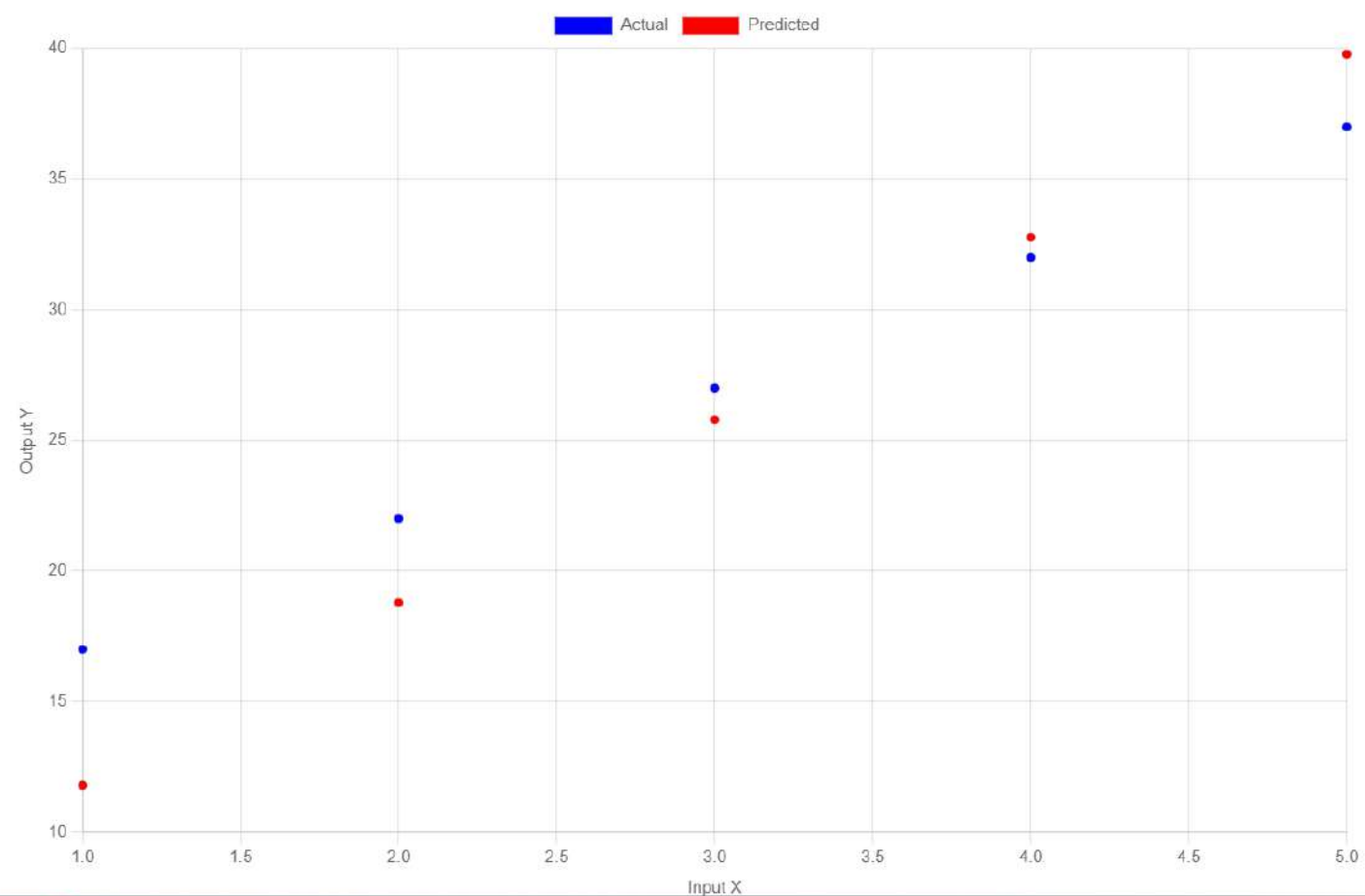
```
Assignment 1: Linear Regression using TensorFlow.js script.js:2
Live reload enabled. index.html:39
Training completed script.js:18
Tensor tfjs@latest:17
[[9.9331627],]
```

>

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Assignment 2: Linear Regression Analysis (TFJS)

Open browser console to see training details



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Epoch 25 | Loss = 238.9846 script.js:20

Epoch 50 | Loss = 84.1197 script.js:20

Epoch 75 | Loss = 37.1497 script.js:20

Training with Learning Rate = 0.01

Epoch 0 | Loss = 615.7687 script.js:20

Epoch 25 | Loss = 15.6870 script.js:20

Epoch 50 | Loss = 13.2427 script.js:20

Epoch 75 | Loss = 11.1798 script.js:20

Training with Learning Rate = 0.1

Epoch 0 | Loss = 702.3038 script.js:20

Epoch 25 | Loss = 4080872704.0000 script.js:20

Epoch 50 | Loss = 24339697778032640.0000 script.js:20

Epoch 75 | Loss = 1.4517032742535325e+23 script.js:20

Predicted vs Actual:

Actual: 17, Predicted: 11.79 script.js:48

Actual: 22, Predicted: 18.79 script.js:48

Actual: 27, Predicted: 25.78 script.js:48

Actual: 32, Predicted: 32.78 script.js:48

Actual: 37, Predicted: 39.77 script.js:48

Unseen Input Prediction:

Input: 10 → Predicted: 74.74 script.js:56

Input: 12 → Predicted: 88.73 script.js:57

Assignment 3: Linear Regression Model Evaluation

Open browser console to see output

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Assignment 3: Model Evaluation using TFJS script.js:2

Live reload enabled. index.html:39

Training completed script.js:18

Predicted vs Expected vs Error: script.js:28

Input: 6, Predicted: 37.36, Expected: 34, Error: 3.36 script.js:32

Input: 7, Predicted: 42.76, Expected: 38, Error: 4.76 script.js:32

Input: 8, Predicted: 48.17, Expected: 42, Error: 6.17 script.js:32

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